

Question Paper

Exam Date & Time: 22-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

PRINCIPLES OF DATA COMMUNICATION [IT_2124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Suppose a transmitter produces 50 W of power. (5)
- A) a. Express the transmit power in units of dBm and dBW.
b. If the transmitter's power is applied to a unity gain antenna with a 900-MHz carrier frequency, calculate the received power in dBm at a free space distance of 100 m.
c. Repeat (b) for a distance of 10 km.
d. Repeat (c) but assume a receiver antenna gain of 2.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Analyze the concept of frame synchronization in TDM. Explain why frame synchronization is necessary and discuss how it ensures that each user receives its allocated time slots in a TDM system. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Analyze the role of the Data Link Layer in the OSI model during data transmission in a Local Area Network. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) Encode the binary word 100011 using an even-parity Hamming code. (5)

- A) Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A satellite communication channel has a bandwidth of 2 MHz and a signal-to-noise ratio (SNR) of 15 dB. (3)

1. Calculate the maximum achievable data rate (channel capacity) according to Shannon's theorem.
2. If the bandwidth is doubled to 4 MHz while the SNR is 15 dB, determine the new channel capacity.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) An FDM system has a total bandwidth of 40 MHz and must support five users transmitting analog voice channels each of 6 MHz bandwidth. (2)

1. Determine how much bandwidth remains for guard bands.
2. If equal guard bands are placed between channels, find their individual widths.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3) Compare Asynchronous Balanced Mode (ABM) and Normal Response Mode (NRM) in HDLC. (5)
Analyze which mode is more suitable for a peer-to-peer communication system where both devices have equal control over the data link.
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A synchronous transmission system transmits data at 1200 bits per second (bps) with a modulation scheme where each symbol represents 4 bits. Calculate the baud rate of the transmission system. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Calculate the baud rate and bandwidth for a signal transmitting at 12 Mbps, The value of $r = 0.5$ (for QPSK). Assume $L = 2$. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4) In a network consisting of six LANs (LAN A, LAN B, LAN C, LAN D, LAN E, and LAN F) connected by five bridges (B1, B2, B3, B4, and B5), the network administrator has implemented the Spanning Tree Protocol (STP) to eliminate loops and maintain efficient communication. After a network expansion, additional redundant connections were added, causing some bridges to be excluded from the active spanning tree. (5)
- A)

1. Explain how the Spanning Tree Protocol (STP) identifies and prevents loops in a network with multiple interconnected bridges.
2. Illustrate how the spanning tree topology reconfigures automatically when Bridge B2 fails, ensuring continued network connectivity.
3. Determine which bridges and links serve as redundant paths, and discuss how these redundancies enhance network fault tolerance and reliability.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A data stream "1100101100" is encoded using Manchester encoding. Calculate the encoded bit sequence. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Compare FDM and TDM in terms of: (2)
1. Bandwidth utilization
 2. Synchronization requirements

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5) Examine the benefits of using CSMA/CD in a local area network (LAN), particularly in terms of reducing the chances of collisions as compared to other random access protocols. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A network administrator is trying to decide between using Selective Repeat ARQ and Go-Back-N ARQ for a system that frequently experiences transmission errors. (3)

1. Compare the efficiency of Go-Back-N ARQ and Selective Repeat ARQ in environments with high error rates.
2. Evaluate the trade-offs between the buffer requirements and throughput for both protocols.
3. Recommend the most suitable protocol for an environment with high transmission errors and explain why it is better suited for this scenario.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Demonstrate the impact of signal encoding on data communication. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_2124:PRINCIPLES OF DATA COMMUNICATION

#	Course Outcome	Description	Marks
1	1	CO 1	13
2	2	CO 2	8
3	3	CO 3	13
4	4	CO 4	12
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_2124:PRINCIPLES OF DATA COMMUNICATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	12	40(80%)
4	Analysis	3	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)